

1. A material manufacturer has to decide how much of each of two types of cloth to produce.
 Each metre of cloth A requires 2 kg of wool, $\frac{1}{2}$ litre of dye, 5 minutes of loom time, and 4 minutes of worker time. Each metre produces a profit of £3.
 Each metre of cloth B requires 1 kg of wool, $\frac{1}{3}$ litre of dye, 4 minutes of loom time, and 5 minutes of worker time. Each metre produces a profit of £2.50.
 The manufacturer has available 6 hours of loom time and 6 hours of worker time for the rest of the day. There are 100 kg of wool and 28 litres of dye.
 - a. Show that the constraint on loom time is redundant.
 - b. Find out how much of each cloth type the manufacturer should produce to maximise her profit. How much profit does she make?
2. An oil company has two refineries.
 Refinery 1 produces 100 barrels of high grade oil, 200 barrels of medium grade oil, and 300 barrels of low grade oil per day. It costs £5000 per day to operate.
 Refinery 2 produces 200 barrels of high grade oil, 100 barrels of medium grade oil, and 200 barrels of low grade oil per day. It costs £12000 per day to operate.
 An order is received for 2000 barrels of high grade oil, 2000 barrels of medium grade oil, and 3600 barrels of low grade oil.
 Assume that the refineries can be operated for part-days, with pro-rata costs and volumes of oil produced.
 - a. Plot or sketch a graph to show the feasible region for this problem.
 - b. Find how to use the refineries so as to fulfil the order:
 - i. at minimum cost;
 - ii. in the minimum amount of time.
 - c. How much surplus oil of each type is produced in each of these cases?
 - d. Find how do use the refineries so as to fulfil the order with minimum surplus oil produced.
3. A small art supply company produces paints that can be used on special surfaces. They mix their red, blue, and yellow base paints to create different colours. On a particular day they are going to make orange and purple paints.
 - They have enough red base to make either 30 litres of orange or 25 litres of purple paint.
 - They have enough blue base to make either 60 litres of orange or 20 litres of purple paint.
 - They have enough yellow base to make either 25 litres of orange or 40 litres of purple paint.
 - a. Plot or sketch a graph showing the feasible volumes of orange and purple paint the company can produce.
 - b. The company anticipates sufficient demand to sell everything that they mix, and they will make a profit of £10 per litre on the orange paint, and £15 per litre on the purple paint. How much of each type of paint should the company make in order to maximise their profit?
 - c. Suppose that the profit per litre of orange paint remains £10, but the profit per litre of the purple paint changes to £ p . For what range of values of p will the companies production plan change, and what will the new plans be?
4. A robot can walk at 1.5 ms^{-1} or run at 4 ms^{-1} . When walking it consumes power at 1 joule per meter; when running, at three times this rate. If its batteries are charged to 9000 J, what is the greatest distance it can cover in half an hour, and how can this be achieved?

5. Describe the possible solutions to Question 4 if the running speed of the robot is $v \text{ ms}^{-1}$.

Harder problems, and some puzzles

6. Three-variable problem

You are going to invest £12,000, and there are three different funds across which to spread your cash. The government bonds have a 7% return, the local bank's Credit Default Swaps have an 8% return, and the high-risk peer-to-peer lending scheme has an expected 12% return. To mitigate risk, you decide not to invest any more than £2,000 in the peer-to-peer lending. For tax reasons, you need to invest at least three times as much in the government bonds as in the bank CDS.

Suppose that you invest x thousand pounds in the bonds, y thousand pounds in the CDS, and the rest in the P2P lending.

- Write down three inequalities that must be satisfied by the variables x and y .
- Also write down and simplify the expression whose value you want to maximise.
- Assuming the year-end yields are as expected, what are the optimal investment amounts?

7. Non-linear objective function.

Maximise $P = x^2 + 12xy + 4y^2$ subject to the constraints $x + 2y \leq 4$ and $x, y \geq 0$.

8. Non-linear constraints.

Maximise $P = 3x + y$ subject to the constraint $3x^2 + 2xy + y^2 \leq 9$.